Team Name: Runtime Terror

Use Case Name: Enterprise Knowledge Copilot (RAG + Agentic Workflow)

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Problem understanding and objective

Why is retrieving the right knowledge still a challenge in data-rich enterprises?

1.Problem Statement

Large IT enterprises face ***"Knowledge Fragmentation"*****,** where insights are trapped in ***scattered PDFs, silos, and duplicate tickets***. This lack of unified intelligence forces engineers to ***waste ~20% of their time*** searching, leading to ***operational delays, repetitive errors, and inflated onboarding costs.***

2.Objective & Stakeholders

To build an Agentic Copilot that transforms static data into a reasoning-capable AI Engineer.

- **Engineers:** Instant, source-backed technical solutions.
- **Support Teams:** Massive reduction in L1/L2 ticket volume.
- **Management:** Faster deployment cycles and data-driven talent scaling.

3.The Dire Need

Traditional search is obsolete for modern, complex architectures (Kubernetes/Cloud). As systems scale, the ***"cost of not knowing" grows***. Companies need a solution that doesn't just retrieve text but understands relationships to prevent critical downtime.

4.Social & Organizational Impact

- **Democratizes Expertise:** Empowers junior developers to perform at a senior level via AI-mentorship.
- **Reduces Burnout:** Eliminates information overload and search-fatigue.
- **Inclusive Growth:** Provides high-tier technical guidance to underprivileged talent entering the workforce.

Market size and business impact

How much is inefficient knowledge access really costing enterprises?

The Economic Drain

- **The 30% Productivity Gap:** Knowledge workers ***waste 2.5 hours*** daily searching for fragmented info.
- **The Talent Leak**: IT firms lose **$15K–$20K per employee** annually in lost productivity.
- **Industry Impact:** A staggering $15B annual loss across global IT services due to information silos.

Market Opportunity (2026-2035)

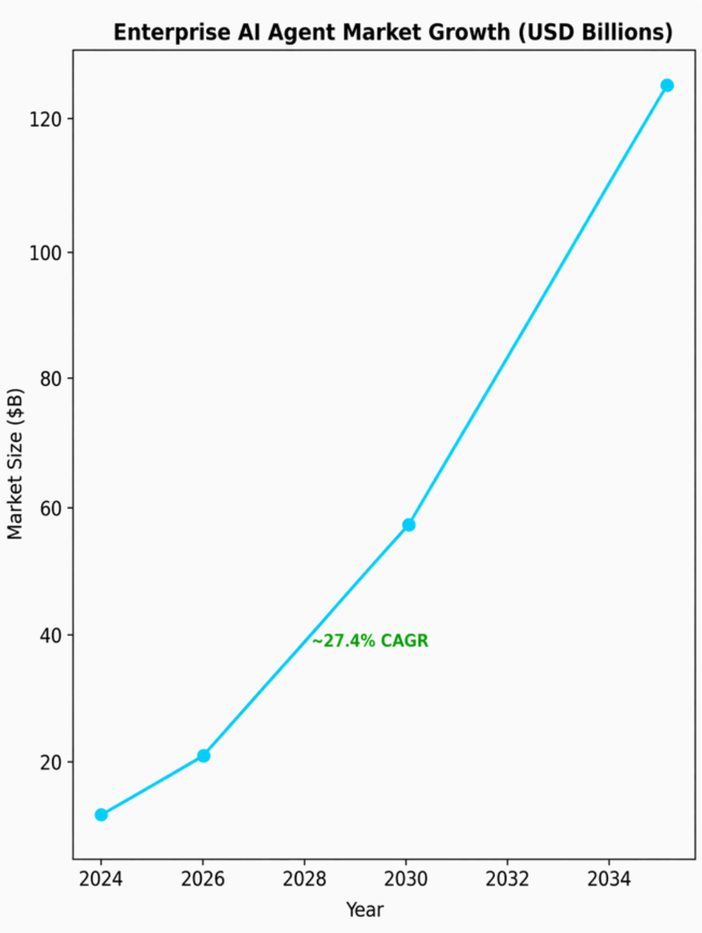
- **The Shift:** Market projected to grow from **$931B to $4.2T** as firms ditch static wikis for AI agents.
- **The CAGR: 27.4% growth** in AI Copilots, signaling a massive enterprise transition.

Our ROI

- **Resolution Speed: 40–60% reduction** in support ticket resolution time.
- **Engineer Onboarding**: Ramp-up time cut from months to days via AI-mentorship.
- Direct Savings: Projected $200K+ annual saving for every 100-engineer team.

Financial Viability

- **Lean Execution:** MVP ready in 4–6 weeks with a minimal PoC investment.
- **Zero Licensing:** Built on **Open Source (LangChain, ChromaDB)** to bypass expensive SaaS fees.



Existing Solutions in the market

Why are $100B Enterprises still struggling with "Search"?

1. Current State & Shortcomings

- **Static Wikis & Tribal Knowledge:** Teams rely on ***"Dead Documentation" (SharePoint/Wikis)*** that is obsolete the moment it's written. Expertise is trapped in unindexed Slack/Teams chats.
- **The "Keyword" Tax:** Traditional search requires users ***to know exactly what to look for***. If you don't use the perfect term, the information remains invisible.
- **Information Sprawl:** ***Knowledge is fragmented*** across PDFs, SOPs, and tickets, leading to a 20–30% productivity drain purely on "information hunting."

2. The Competition & The "Context Gap"

- **MS Copilot / AWS Q:** While powerful, they are ***"Broad but Shallow"***. They lack the deep, private logic of your specific legacy systems and internal "hacks."
- **Standard RAG Tools:** Most tools (Glean/Guru) are ***Passive Chatbots***. They return flat text chunks but lack the Reasoning Layer needed to actually diagnose a system failure.
- **The Hallucination Risk:** Generic bots often ***"guess" technical steps*** when data is missing, creating a dangerous Trust Gap in high-stakes IT environments.

3. Our Differentiators

- **GraphRAG++:** We move from ***"Flat Chunks" to "Relational Logic"***. By using a Knowledge Graph, We understand that Error X in Service A is actually caused by Database B.
- **Self-Learning Feedback Loop:** Our system isn't frozen. It ***"listens"*** to every newly resolved ticket and user "Thumbs Up," evolving its intelligence in real-time.
- **Confidence-Gated Escalation:** We solve the trust issue with a ***"Trust Score"***. If the AI isn't 90% sure, it doesn't guess, it summarizes the case and hands it off to a human expert.

Proposed solution approach

"From static knowledge retrieval → to intelligent reasoning, learning, and action."

Solution Approach & High-Level Design

We utilize a Hybrid Graph-RAG & Agentic Workflow. Instead of simple text retrieval, it models enterprise data as an interconnected Knowledge Graph.

- **Ingestion:** Scrapes and chunks PDFs, Wikis, and Tickets using LlamaParse for high-fidelity table/chart extraction.
- **Orchestration:** Employs a Multi-Agent ReAct pattern (via LangChain/LangGraph) to plan, search, and execute diagnostic tasks.
- **Storage:** Dual-database approach—ChromaDB for vector similarity and Neo4j for relational reasoning.

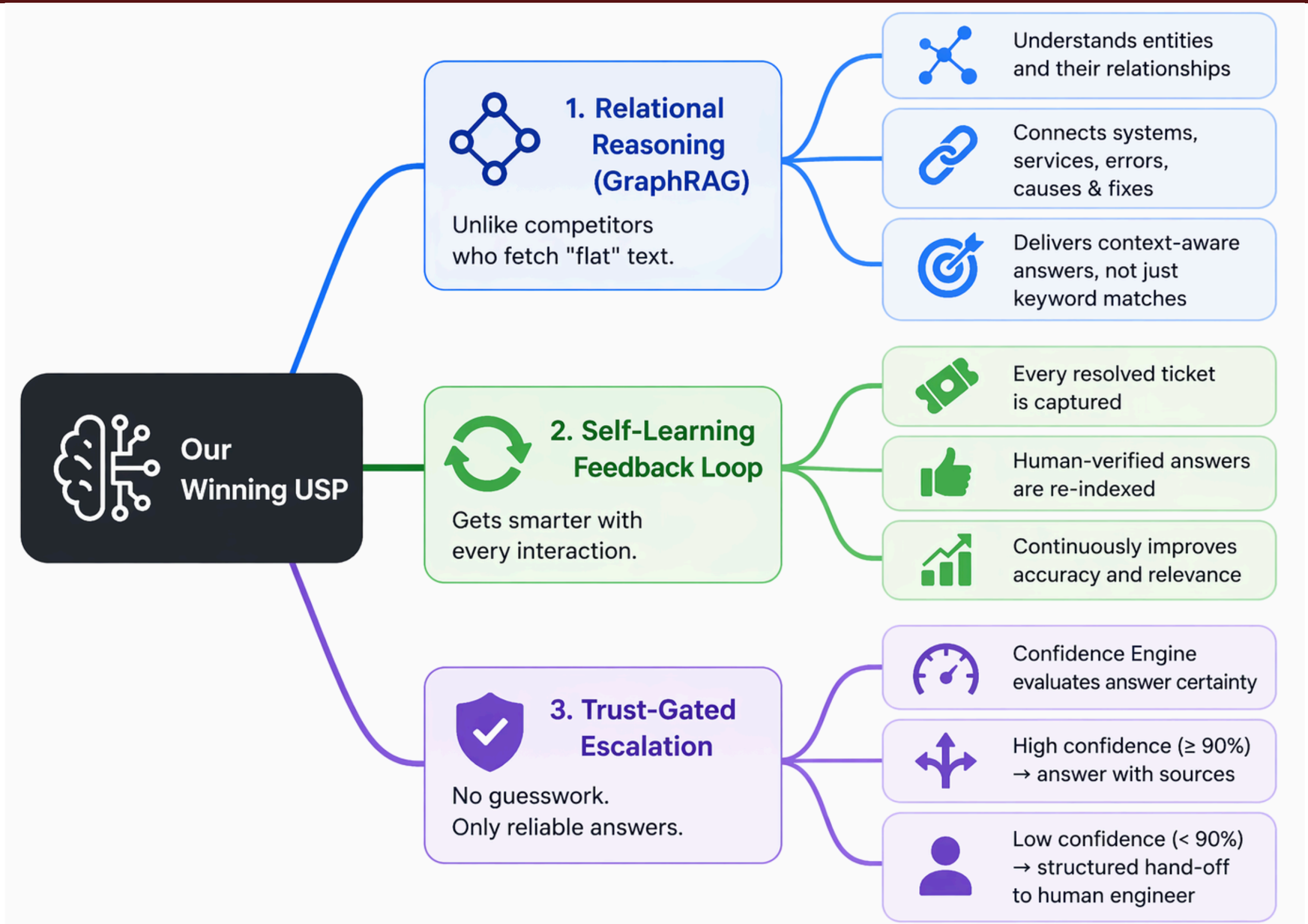
Our Winning USP (Unique Selling Proposition)

- **Relational Reasoning (GraphRAG):** Unlike competitors who fetch "flat" text.
- **The Self-Learning Feedback Loop:** Every resolved ticket and human-verified answer is re-indexed, meaning the system gets smarter with every interaction.
- **Trust-Gated Escalation:** Features a unique Confidence Engine. If the AI is unsure, it drafts a detailed hand-off for a human engineer instead of hallucinating.

Dependencies & Assumptions

- **Data Access:** Assumes secure read-access to internal documentation repositories (Confluence/SharePoint/Jira).
- **Compute:** Requires high-performance LLM API access (GPT-4o/Claude 3.5) or local inference for sensitive data.
- **Human-in-the-Loop:** Effectiveness relies on initial expert verification to "seed" the self-learning loop with high-quality resolutions.

Proposed solution approach



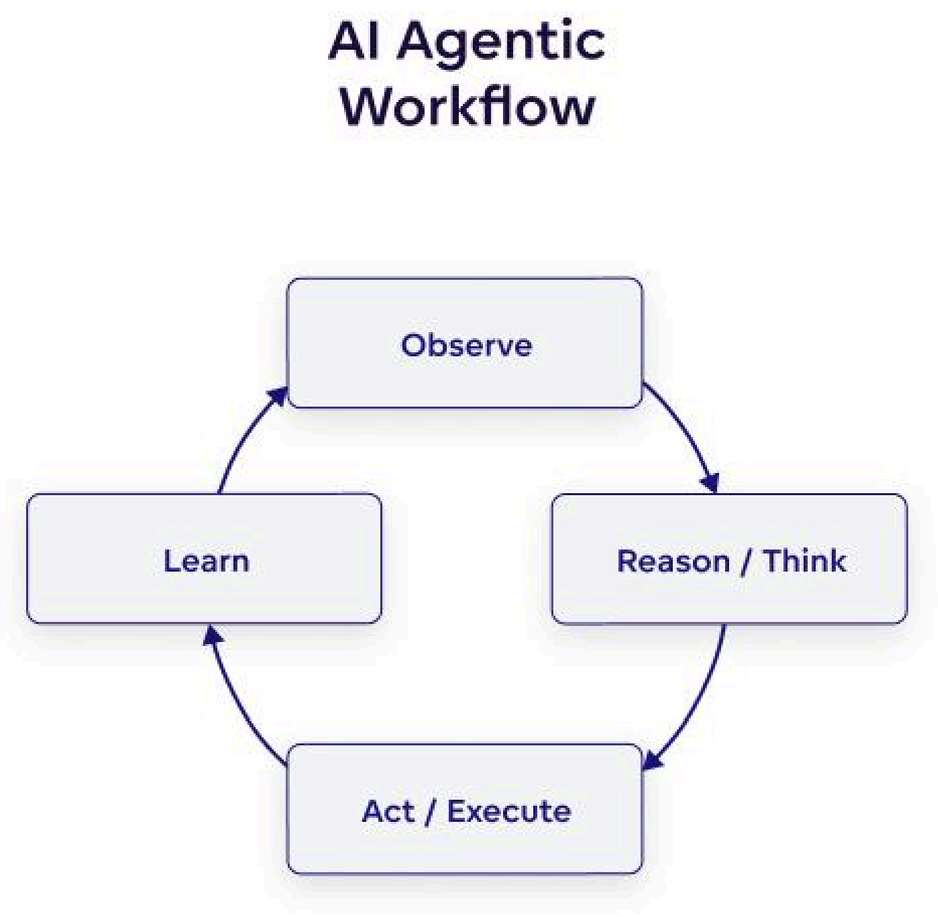
Technical Details

Technical Feasibility

- **Modular Open-Source Stack**: Leveraging ***LangChain and FastAPI*** ensures we aren't "reinventing the wheel." These are industry-standard, well-documented frameworks.
- **Model Agnostic Design:** Our architecture works with OpenAI, Anthropic, or ***local models*** like **Llama 3**, allowing for flexible deployment based on the client's existing infrastructure.
- **Proven RAG Patterns:** We utilize **Hybrid Search (Vector + Keyword)** and Reranking, which are established methods for high-accuracy information retrieval.

Business Viability

- **High ROI / Low OpEx:** By using ***open-source vector databases*** (ChromaDB/Qdrant) and ***graph databases (Neo4j)***, we eliminate recurring per-user licensing costs associated with proprietary SaaS.
- **Reduced MTTR:** The system directly addresses the ***"Time-to-Resolution"*** KPI, which is the primary driver of cost in IT Service Management (ITSM).
- **Seamless Integration:** Designed to **plug into existing enterprise ecosystems** (Slack, Microsoft Teams, Jira) without disrupting established workflows.



Design Considerations: Security & Scalability

1. **Security & Data Integrity**

- **PII Masking:** Automated redaction layers ensure **GDPR** compliance before data reaches the **LLM**.
- **Enterprise RBAC:** Inherits existing permissions; access is restricted based on organizational hierarchy.
- **VPC Deployment:** Option for Private Cloud hosting to **guarantee zero data leakage** to public providers

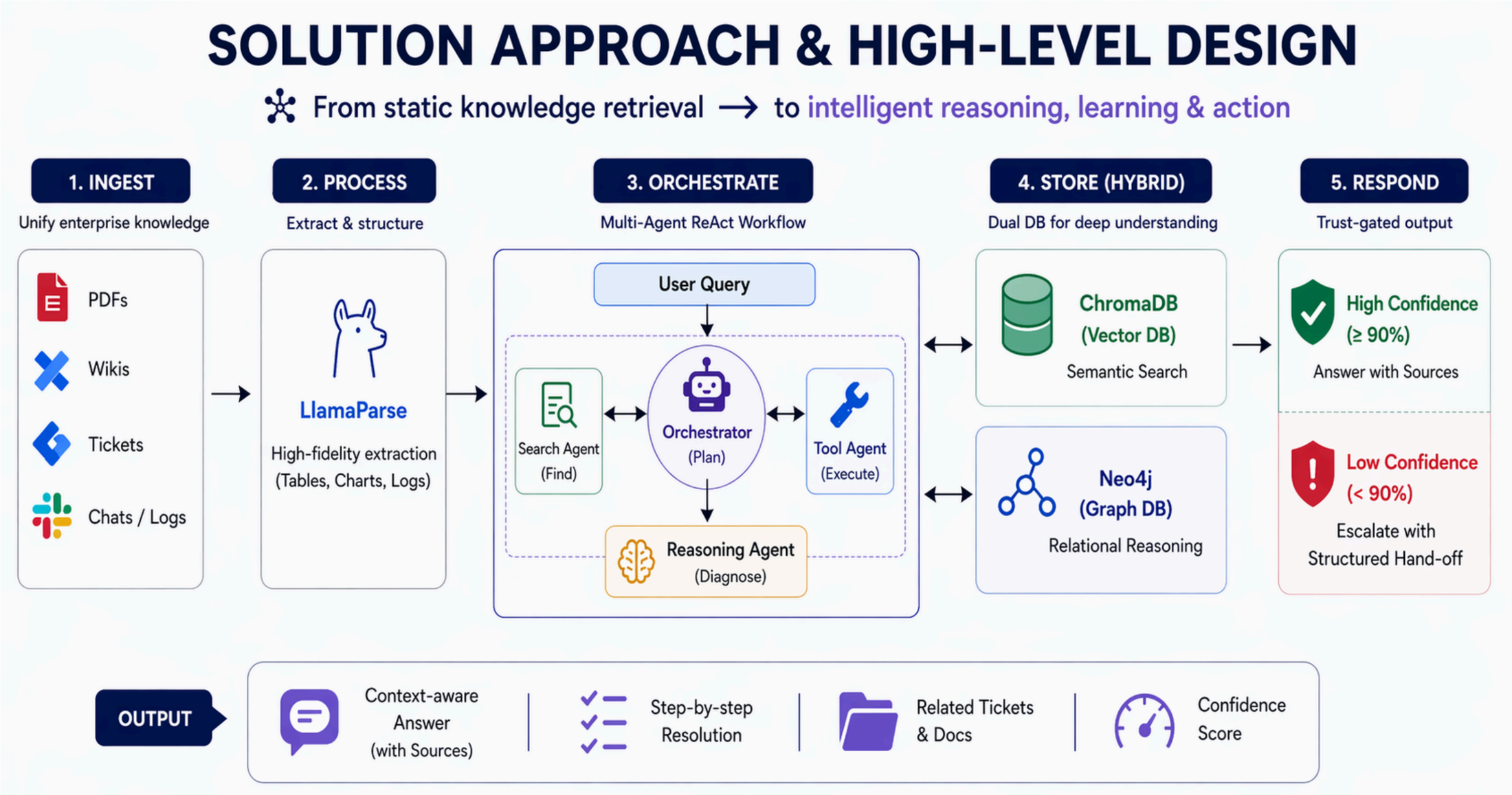
2. **Scalability & Performance**

- **Async Engine:** Powered by **FastAPI & Celery** to handle thousands of concurrent queries without latency.
- **Distributed Vector DB:** Uses **Qdran**t to maintain sub-second **retrieval as data grows** to millions of docs.
- **K8s Ready:** Fully containerized **Microservices (Docker)** for seamless horizontal scaling.

3. **Technical Complexity & Validation: “The Engineering Challenge”**

- **Agentic Orchestration:** Managing complex **multi-step reasoning** and state transitions via **LangGraph.**
- **GraphRAG Synergy:** Synchronizing **Neo4j (Relational)** with **Vector DB (Semantic) for a 360° knowledge view.**
- **Real-time Synchronization:** Ensuring the Knowledge Graph stays updated as new tickets are resolved.

4. **The "Gold Standard" Evaluation**

- **RAGAS Framework:** We move beyond "**vibe-based**" testing to mathematical scoring of **Faithfulness, Relevancy, and Context Precision.**
- **Continuous Monitoring:** **Automated feedback loops** to detect and eliminate hallucinations in production.

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